Tony Xu

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Education

University of Toronto 2022 - 2027
PhD Candidate in Medical Biophysics *Cumulative GPA: 4.0/4.0*

- **Awards**: Google PhD Fellowship (Nov. 2024), Postgraduate Scholarships – Doctoral Program (Apr. 2024), Vector Scholarship in Artificial Intelligence (Sept. 2022), Canada Graduate Scholarships - Master's (Sept. 2022)

University of British Columbia 2017 - 2022
BASc Electrical Engineering, Biomedical specialization, Minor in Physics *Cumulative GPA: 3.92/4.0*

Research Experiences

Graduate Research Assistant Sept. 2022 – Present
UofT Medical Biophysics, supervised by Anne Martel and Maged Goubran *Toronto, ON*

- Working on self-supervised learning (SSL) methods applied to huge 2D WSIs and 3D microscopy images to learn salient image representations from unlabelled data and reduce reliance on expert annotations
- Developed μ3DINO: a foundation model for 3D microscopy and leveraged it in a novel segmentation pipeline that addresses noisy ground truth and hard image regions [1]
- Improved tuning of WSI-level foundation models to leverage multimodal, multitask, and pan-cancer datasets [2]
- Scaled SSL methods to 3D medical images, developed 3DINO: a 3D foundation model for medical imaging [3, 5]
- Developed an interpretable pipeline for lymph node metastasis detection in WSIs [4], working to design output format to enhance clinican trust, collaborating with hospital towards deployment in pathologist workflows
- Mentored 5+ undergraduate/master's students on deep learning research projects applied to medical imaging, provided guidance on technical implementation, research direction, and result analysis

Computational Scientist Intern Jun. 2022 – Sept. 2022
Samsung Research America – Think Tank Team *Mountain View, CA*

- Ideated, researched, and designed experiments to push the frontier of Samsung technology
- Implemented efficient signal processing, computer vision detection and SLaM algorithms to prototype products

Undergraduate Research Student May 2019 – May 2021
Sunnybrook Research Institute, supervised by Anne Martel *Toronto, ON*

- Investigated SSL approaches applied to histopathology image data [6]
- Used DL and computer vision techniques to analyse breast cancer WSIs for cancer detection with PyTorch [7]
- Developed an efficient SSL technique using domain-specific double pretraining on a variety of SSL methods [8]

Relevant Publications

[1] **Harnessing SSL for Segmentation in 3D Microscopy with Noisy Labels and Hard Patches** 2026
T. Xu, A. Attarpour, S. Patel, F. Yu, M. Rozak, B. Stefanovic, A. L. Martel, M. Goubran
ECCV 2026, Malmö

[2] **ModalTune: Fine-Tuning Slide-Level Foundation Models with Multi-Modal Information for Multi-task Learning in Digital Pathology** 2025
T. Xu, V. Ramanathan*, P. Pati, A. Faruk, M. Goubran, A. L. Martel*
ICCV 2025, Honolulu

[3] **A generalizable 3D framework and model for self-supervised learning in medical imaging** 2025
T. Xu, S. Hosseini, C. Anderson, A. Rinaldi, R. G. Krishnan, A. L. Martel, M. Goubran
npj Digital Medicine

[4] **AI-assisted detection of breast cancer lymph node metastases in the post-neoadjuvant treatment setting** 2025
T. Xu, D. Bassiouny, C. Srinidhi, M. S. W. Lam, M. Goubran, S. Nofech-Mozes, A. L. Martel
Laboratory Investigations

[5] **Masked image modeling for label-efficient segmentation in two-photon excitation microscopy** 2023
T. Xu, M. Rozak, E. Ntiri, A. Dorr, J. Mester, B. Stefanovic, A. L. Martel, M. Goubran
Medical Image Learning with Limited and Noisy Data at MICCAI 2023, Vancouver

[6] **Self supervised contrastive learning for digital histopathology** 2022
O. Ciga, T. Xu, and A. L. Martel
Machine Learning with Applications

[7] **Overcoming the limitations of patch-based learning to detect cancer in whole slide images** 2021
O. Ciga, T. Xu, S. Nofech-Mozes, S. Noy, F. I. Lu, and A. L. Martel
Scientific Reports

Preprints

[8] **Resource and data efficient self supervised learning** 2021
T. Xu, O. Ciga*, and A. L. Martel*
arXiv preprint

Professional Experiences

Software Development Engineering Intern Jun. 2021 – Aug. 2021
Amazon *Vancouver, BC*
- Designed and built an operational dashboard to consolidate and securely manage team workflows
- Deployed solution using AWS Services (Lambda Step Functions, CDK, DynamoDB, S3, CloudFront, etc.)
- Created an authenticated UI using React/TypeScript, infrastructure code using AWS CDK in TypeScript, and implemented security measures using AWS Cognito to properly secure access of confidential data

Machine Learning Engineering Intern May 2020 – Jan. 2021
Flex Artificial Intelligence Inc. *Vancouver, BC*
- Developed an end-to-end patented computer vision pipeline to detect fine-grained form errors in real world exercise videos, such as raising toes during deadlift, to provide insight on a user's exercise performance
- Researched and implemented approaches to analyse video data, including pose detection, spatio-temporal attention, self-supervised temporal alignment, deformable convolutions, and triplet models for anomaly detection
- Improved frame-level error classification accuracy by 17.3% with fully completed pipeline

Selected Projects

FLARE 2025 Challenge Task 4 Jun. 2025 – Sept. 2025
- 1st place on FLARE 2025 Task 4: Foundation Models for 3D CT and MRI, presented at MICCAI 2025 workshop
- Formulated 3D SSL algorithm paired with LoRA to train multimodal foundation model for CT and MRI images
- Improved performance on diverse set of downstream tasks in medical imaging (segmentation, classification, survival prediction, regression, etc.)

SELMA3D 2024 and 2025 Challenges Sept. 2024 – Sept. 2025
- 1st and 3rd place on SELMA3D 2025 and 2024: challenges on using SSL to improve segmentation of 3D light-sheet microscopy images, presented algorithm at MICCAI 2024 and 2025 workshop
- Implemented novel 3D SSL and segmentation pipeline integrating 3DINO, nnUNet and LoRA frameworks
- Used SSL representations to curate unlabelled dataset and pseudolabelled to improve out-of-domain generalization

Diffusion-Based MRI Subsampled K-Space Reconstruction Dec. 2023 – Feb. 2024
- Used DDNM to formulate MRI K-space subsampling as inverse problem
- Trained 2D and 3D diffusion models to perform zero-shot MRI reconstruction from arbitrary K-space subsampling patterns (including non-Cartesian patterns using nonuniform FFT)
- Achieved strong results on various 2D and 3D MRI datasets, conference paper in progress

SemiCOL Challenge Feb. 2023 – Jun. 2023
- 2nd place on arms 1 and 2 in SemiCOL: challenge using semi-supervised learning for colorectal cancer detection, presented algorithm at ECDP 2023: the 19th European Congress on Digital Pathology
- Created efficient tissue segmentation pipeline using an ensemble of UNet and UNETR models respectively using PAWS and DINO semi- and self-supervised pretrained ResNet50 and ViT encoders

Technical Skills

Languages: Python, JavaScript/TypeScript, C, SQL, MATLAB
Deep Learning: PyTorch, Keras/TensorFlow, scikit-learn, OpenCV, Gym
Tools and Frameworks: Git, Bash/Linux, Docker, AWS, React, HPC